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Dataset:- Seeds classification Dataset with 3 classes and numerical features, so it is suitable for a 4-2-3 neural network.

classes: kama
Rosa
Canadian

4 features: $x_1 = \text{Area}$
 $x_2 = \text{Perimeter}$
 $x_3 = \text{Kernel Length}$
 $x_4 = \text{kernel width.}$

Input Layer $\rightarrow$ 4 neurons
 $x = [x_1, x_2, x_3, x_4]$

first row $x = [15.26, 14.84, 5.763, 3.312]$

weights for H_1 $w_1 = 0.08, w_2 = -0.05, w_3 = 0.10, w_4 = 0.06$

H_2 $w_5 = -0.03, w_6 = 0.06, w_7 = 0.08, w_8 = -0.04.$

Bias $B_1 = -0.5, B_2 = 0.2.$

Calculate H_1 $Z_{H_1} = x_1 w_1 + x_2 w_2 + x_3 w_3 + x_4 w_4 + B_1$

$$= (15.26 \times 0.08) + (14.84 \times -0.05) + (5.763 \times 0.10) + (3.312 \times -0.06) - 0.5$$

$$= 1.2208 - 0.742 + 0.5763 + 0.19872 - 0.5$$

$$\boxed{Z_{H_1} = 0.75382}$$

Sigmoid for $H_1 \Rightarrow H_1 = \frac{1}{1 + e^{-Z_{H_1}}} \Rightarrow \frac{1}{1 + e^{-0.75382}} \Rightarrow \boxed{H_1 = 0.6800}$

Calculate H_2 $Z_{H_2} = x_1 w_5 + x_2 w_6 + x_3 w_7 + x_4 w_8 + B_2$

$$= (15.26 \times -0.03) + (14.84 \times 0.06) + (5.763 \times 0.08) + (3.312 \times -0.04) - 0.2$$

$$= 0.4578 + 0.8904 + 0.46104 - 0.13248 - 0.2$$

$$\boxed{Z_{H_2} = 0.56116}$$

Sigmoid for $H_2 \Rightarrow H_2 = \frac{1}{1 + e^{-Z_{H_2}}} = \frac{1}{1 + e^{-0.56116}}$

$$\boxed{H_2 = 0.6367}$$

$$\Rightarrow H = [0.6800, 0.6367]$$

Output layer.

Kama $w_9 = 1, w_{10} = 0.5, B = -0.1$
Rosa $w_{11} = -0.5, w_{12} = 1, B = 0.05$
Canadian $w_{13} = 0.2, w_{14} = -0.1, B = -0.05$

for kama $z = (H_1 w_9) + (H_2 w_{10}) - 0.1$
 $= (0.6800 \times 1) + (0.6367 \times 0.5) - 0.1$
 $= 0.6800 + 0.31835 - 0.1$
 $z_{\text{kama}} = 0.89835$

for Rosa $z = (H_1 w_{11}) + (H_2 w_{12}) - 0.05$
 $= (0.6800 \times -0.5) + (0.6367 \times 1) - 0.05$
 $= -0.3400 + 0.6367 - 0.05$
 $z_{\text{Rosa}} = 0.2467$

for Canadian $z = (H_1 w_{13}) + (H_2 w_{14}) - 0.05$
 $= (0.6800 \times 0.2) + (0.6367 \times -0.1) - 0.05$
 $= 0.1360 - 0.06367 - 0.05$
 $z_{\text{Canadian}} = 0.02233$

Output $= [0.89835, 0.24670, 0.02233]$

High Score
↓
→ 0.89835

predicted class = kama

final Answer ↑

$$B_1 = -0.5 \rightarrow H_1$$

$$B_2 = -0.2 \rightarrow H_2$$

